Chapter

Introductory Chapter: Unveiling Present and Future Horizons of Open Source

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1. Introduction

The roots of the Open Source movement can be traced back to the early days of computing [1], with the advent of freely shared software among the pioneering community of programmers. However, it was not until the 1980s that the term "open source" was coined and popularized: a group of developers led by Linus Torvalds created the Linux operating system, which was released as free and open source software. This project paved the way for the Open Source movement by demonstrating the power and potential of collaborative software development [2]. The Open Source movement continued to gain traction in the enterprise world in the early 2000s, with companies like IBM, Red Hat, and Oracle offering open source solutions for business applications, databases, and operating systems [2]. Today, Open Source lives a new revolution, in which the cornerstone role of software in societal advance is being embraced as well in the scientific arena, and has paired the term with others like Open Data, Open Access, and Open Science.

1.1 The tapestry of the open movements

The practical benefits of collaborative work and exploration are without discussion [3]. Indeed, they extend beyond software, giving rise to parallel movements in various domains. First, the Open Data [4] movement emerged, claiming for the unrestricted availability and sharing of data. Recognizing the transformative power of data in driving innovation, transparency, and accountability, this movement has influenced governments, research institutions, and businesses to adopt open data practices. It also addresses the empowerment of citizens toward gaining control over their own data.

In tandem, Open Access [5] appeared, another consequential offshoot, which focuses on removing barriers to scholarly publications. Scholars and researchers advocate for unrestricted access to academic literature, ensuring that knowledge is not confined by paywalls. Open Access has redefined the dissemination of research findings, fostering a more egalitarian access to information.

This path has led to Open Science [6], a philosophy advocating for the transparent and collaborative sharing of scientific knowledge, thus aligning closely with the ethos of the Open Source movement. The goal is to make scientific research more accessible, reproducible, and inclusive. Collaborative platforms, open-access journals, and initiatives promoting data sharing have become integral components of the Open Science

movement, reshaping the landscape of academic research. Institutions like the European Union have reformulated their funding programs to adopt strong Open Science commitments [7].

These interconnected movements, often referred to collectively as "the Open Movements", form a collaborative tapestry that extends beyond the realm of technology. As technology, science, data, and knowledge intertwine, the Open Movements collectively shape a paradigm that emphasizes collaboration, inclusivity, and the democratization of information. The success of one movement often paves the way for advancements in others, creating a synergistic ecosystem that has become a hallmark of progressive and inclusive practices in the XXI century. The evolution of these movements reflects a commitment to openness and underscores the transformative power of collaborative efforts in shaping the future of our interconnected world.

2. Purpose and goals

In editing this book, a clear mission was set: to provide a state-of-the-art look at the present and future of Open Source. This is not a technical manual; it is a mosaic of perspectives that examines Open Source from various vantage points. We aim to illuminate its critical role in Open Science, its strategic significance for small businesses, its democratic potential within public administrations, and the nuanced implications revealed through a societal-focused analysis. We also pursued the exploration of its imminent coexistence with the emergent large AI-powered systems.

Our ultimate purpose is twofold: to provide a sneak peek of the fringe of Open Source, and to present a forward-looking panorama of its far-reaching implications from a comprehensive perspective, to foster critical thinking about the societal impact of the Open Movements. By doing so, we invite readers to adopt a holistic viewpoint, understanding the intricate threads that weave the fabric of Open Source into the broader societal canvas.

3. Conclusion

At its core, this book goes far beyond lines of code and celebrates the spirit of collaboration that defines the Open Source community. It serves as a testament to the collective ingenuity that has propelled Open Source to the forefront of technological progress. Through case studies and real-world examples, we showcase the accomplishments and potential of a community driven by a shared vision for a more open, accessible, and inclusive digital future.

Whether the reader is a seasoned technologist, a business strategist, a public servant, or an enthusiast curious about the digital realm, this volume hopes to be an inspiring resource. The Open Source phenomenon is not confined to the realm of technology—it transcends its technological roots and has become a cultural, societal, and transformative force. By offering diverse perspectives, we hope to inspire further thinkers, builders, and advocates who will shape the trajectory of open source in the years to come.

With "Open-Source Horizons - Challenges and Opportunities for Collaboration and Innovation", we extend an invitation to engage in an exploration, where the meticulous examination of collaborative innovation converges with intellectual rigor to unravel the complexities and prospectives of Open Source.

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References

[1] Brasseur VM. Forge your Future with Open Source. Raleigh, North Carolina (US): Pragmatic Programmers; 2018. ISBN 978-1-68050-301-2

[2] Michels M. A Brief History of Open Source. 2021. Available from: https://maximilianmichels.com/2021/history-of-open-source/ [Accessed: November 13, 2023]

[3] Kilamo T, Hammouda I, Chatti MA. Teaching collaborative software development: A case study. In: 34th International Conference on Software Engineering (ICSE). Zurich, Switzerland: IEEE; 2012. pp. 1165-1174. DOI: 10.5555/2337223.2337376

[4] Open Data Handbook. Available from: https://opendatahandbook.org/guide/en/what-is-open-data/ [Accessed: November 13, 2023]

[5] Dutch National Open Access Website: What is open access? Available from: https://www.openaccess.nl/en/what-is-open-access [Accessed: November 13, 2023]

[6] UNESCO. Open Science: Making Science more Accessible, Inclusive and Equitable for the Benefit of All. Paris, France: UNESCO; 2019. Available from: https://www.unesco.org/en/open-science [Accessed: November 13, 2023]

[7] European Commission. The EU's Open Science Policy. Brussels, Belgium: European Commission; 2019. Available from: https://research-and-innovation.ec.europa.eu/strategy/strategy-2020-2024/our-digital-future/open-science_en [Accessed: November 11, 2023]